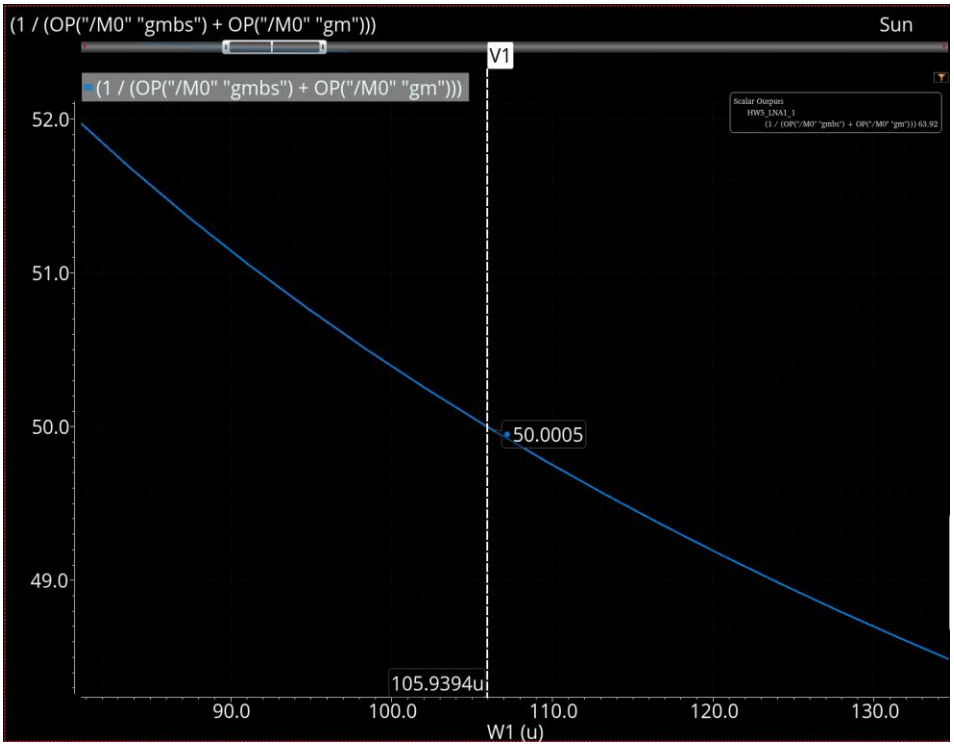
ECE6420 Homework 5 Solution

(Fall 2026)

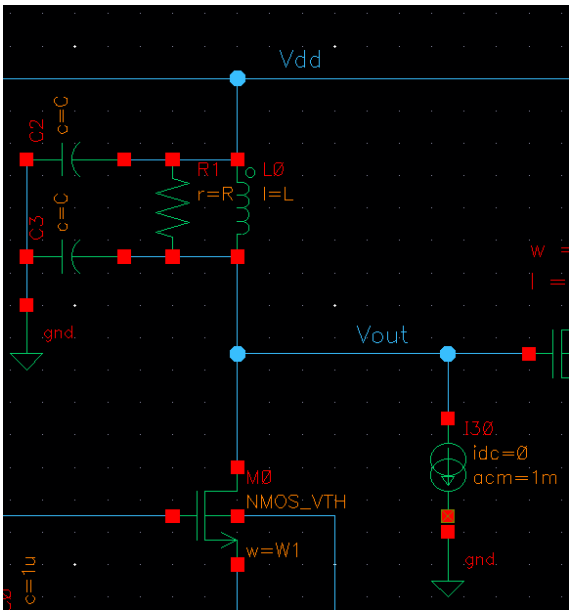
(a). Consider the common gate circuit shown on the right, where M2 represents a typical load capacitance for now. Assuming λ=0, find the width of M1 such that g_m1+g_mb1=(50Ω)^(-1).

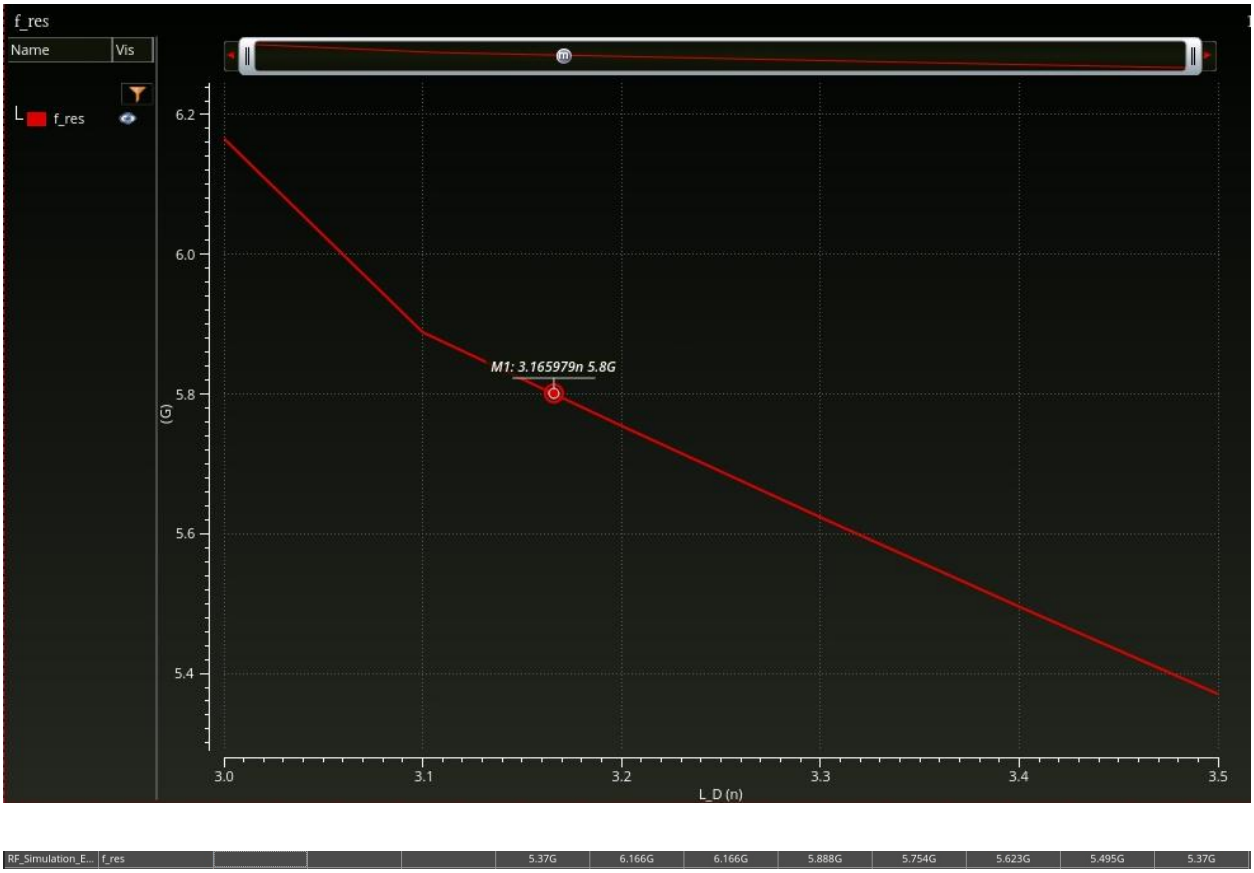
By doing a parameter sweep on the size of the transistors the new width found to be 105.94 um, resulting in 50.0005 Ohm resistance.



(b). Use simulations to determine the value of L1 for the resonance at 5.8 GHz. The inductor must be modeled as shown, where Rp is chosen to give a Q of 8 at the frequency of interest and Cp=10fF for every nanohenry of inductance.

We can set up the simulation like this to check the resonant frequency:





Using simulation the estimated value for L turned out to be 3.15 nH. This was achieved by sweeping the values of L.

$$R_P = Q\omega L = \ 8 \times 2\pi \times 5.8 \times 3.15 = 918\ Ohm$$

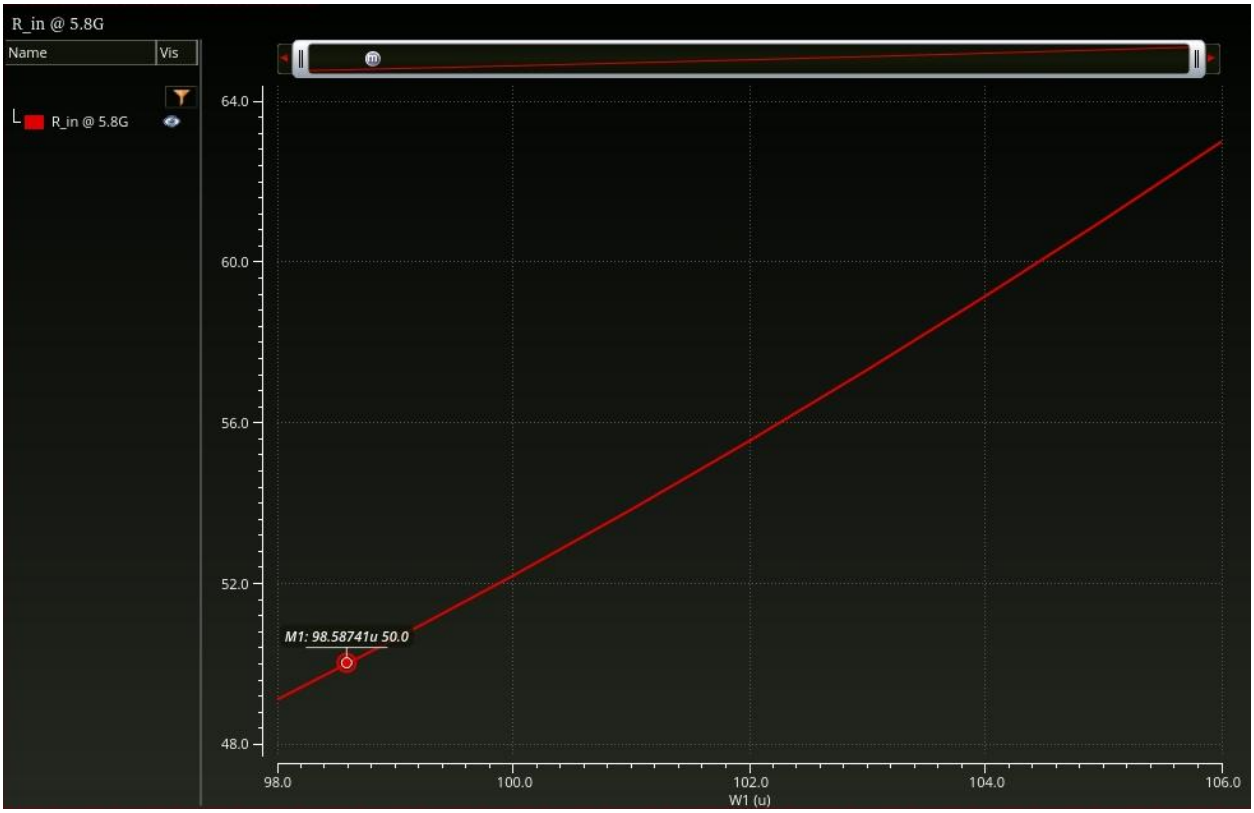
$$C_P = L \times 10^{-5} F = \ 31.5\ fF$$

(c). Now find the input resistance (i.e., the real part of the input impedance) by simulations and explain why it is not equal to 50Ω. Adjust the width of M1 to obtain a 50Ω input resistance again.

In part (a), it was found that W1=106 um leads to gm+gmb=(50ohm)$^{-1}$. However, the real part of input impedance turns out to be 63ohm. (Add an AC source to the input and its voltage is equal to the impedance.)

Test	Output	Nominal
Filter	Filter	Filter
RF_Simulation_E...	/M2/D	
RF_Simulation_E...	/M1/D	
RF_Simulation_E...	v /Vin; ac mag(V)	
RF_Simulation_E...	(gm+gmb)^-1	50.02
RF_Simulation_E...	R_in @ 5.8G	63

W1 that results in 50ohm input resistance is 98.5um as shown below.



RF_Simulation_E...	(gm+gmb)^-1			50.02	50.58	50.58	50.51	50.43	50.36	50.3	50.23	50.16	50.09	50.02
RF_Simulation_E...	R_in @ 5.8G			49.11	63	49.11	50.62	52.2	53.84	55.55	57.32	59.15	61.05	63

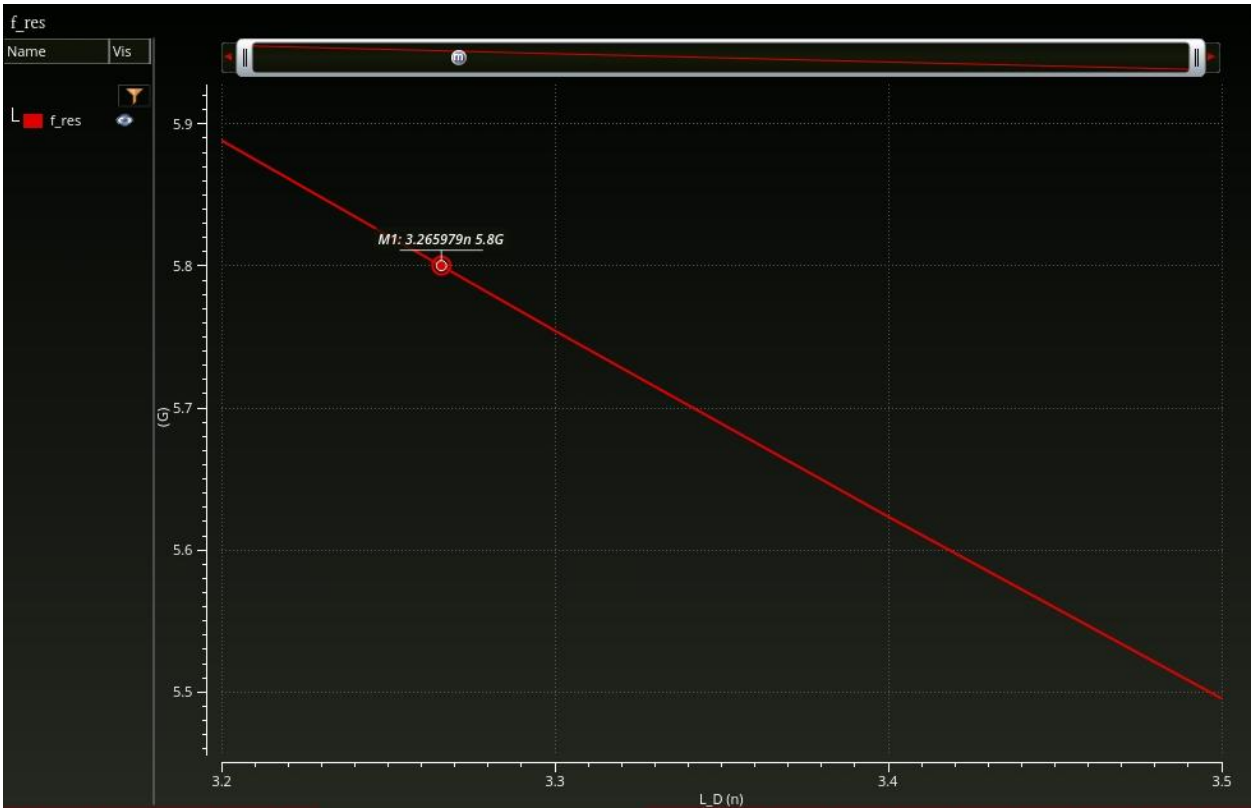
By checking the expression of R_{in}:

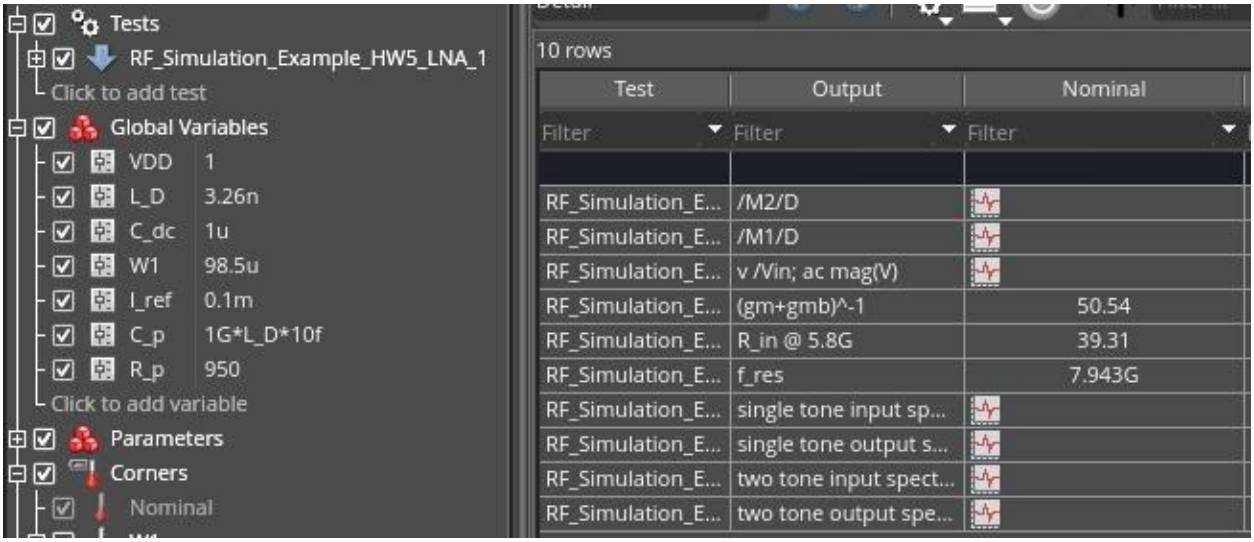
$$R_{in} = \frac{r_0 + Z_p}{1 + (g_m + g_{mb})r_0}$$

(Z_p is the impedance of the drain RLC tank) This is primarily due to 2 reasons:

1. The channel length modulation (r_0): it's a finite value.
2. Contribution of the Z_p.

The resonance frequency is shifted since W1 is smaller. To shift back the resonance frequency to 5.8 GHz we need to change L to 3.26 nH as shown in the following figure. Then, Rp=950 ohm.

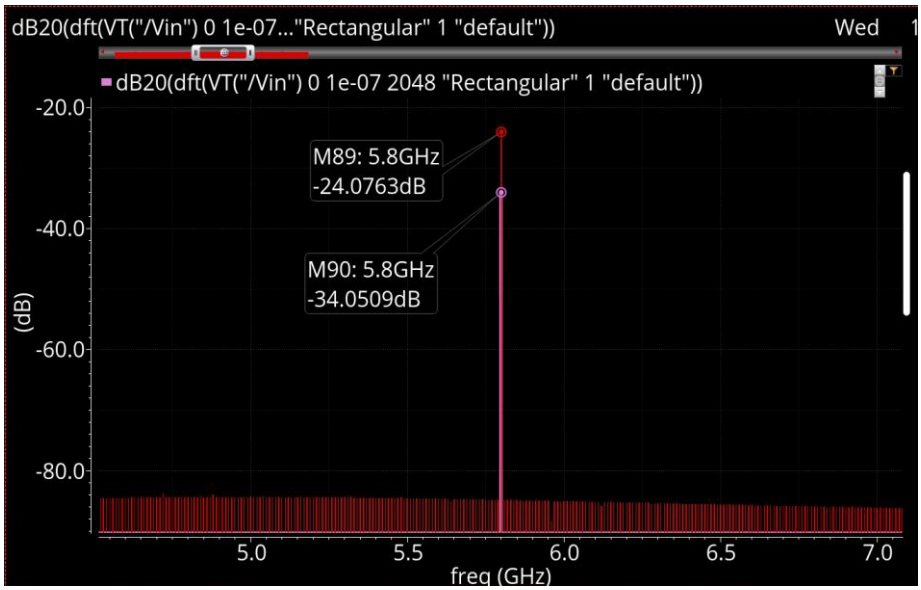


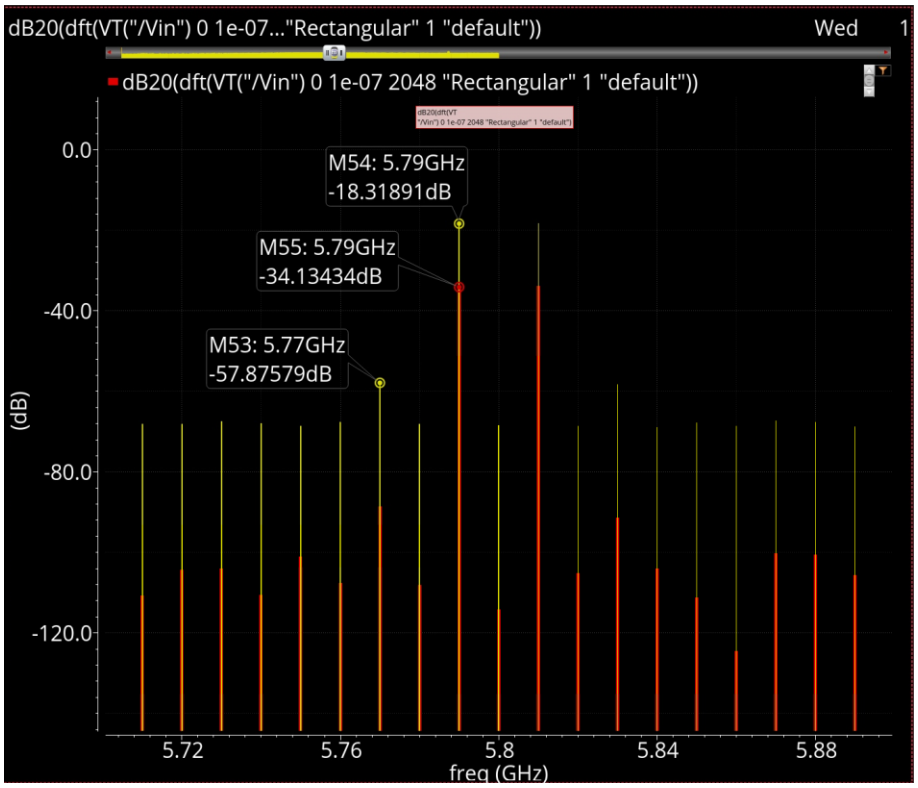


Final results are shown as above. Changing L1 to retune the output shifted Rin to 39.31 ohm.

(d). Using the shortcut method (two tone test) to compute the IIP3 and voltage gain of the circuit at 5.8 GHz.

Two tones at 5.79 GHz and 5.81 GHz with 20 mV amplitudes are applied to the input and by performing DFT, the input/output voltage spectrum was plotted as below. All plots are on a dB20 scale.





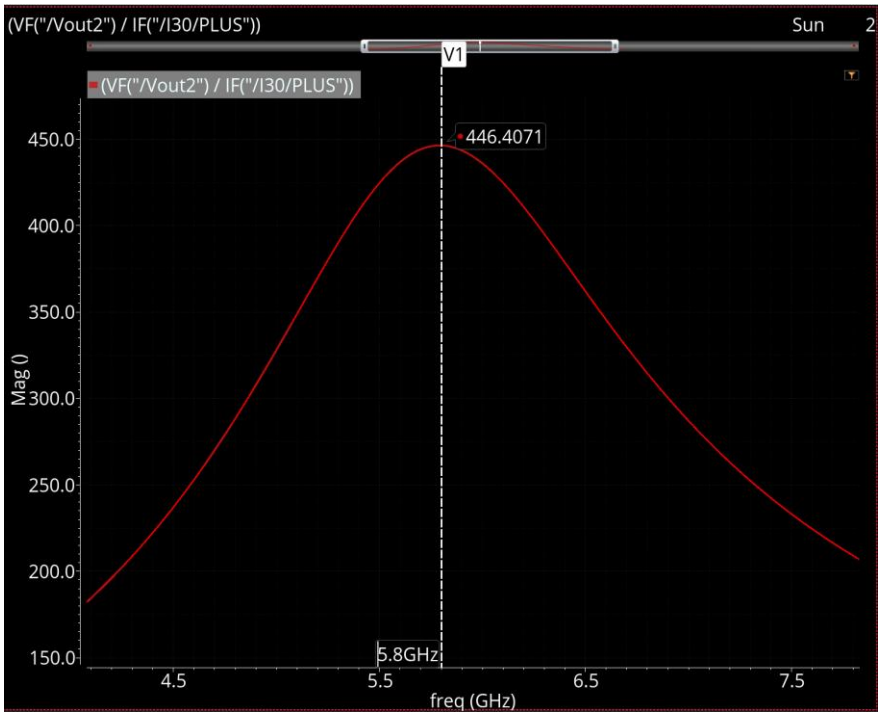
Output: -18.31 dBV = -5.3dBm; Input: 20mV = -34 dBV; IM3: -57.85 dBV = -44.8dBm

$$G = -24.07dBV - (-34.05dBV) = 9.98\ dB$$

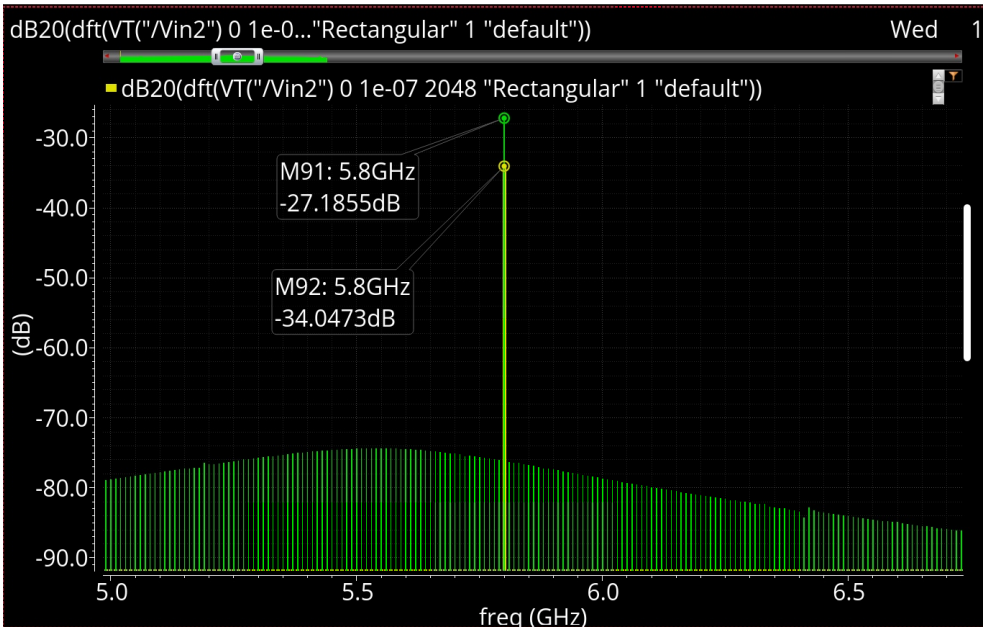
$$IIP3 = \frac{\Delta P}{2} + P_{in} = \frac{-5.3 + 44.8}{2} - 34 = -14.25\ dBV$$

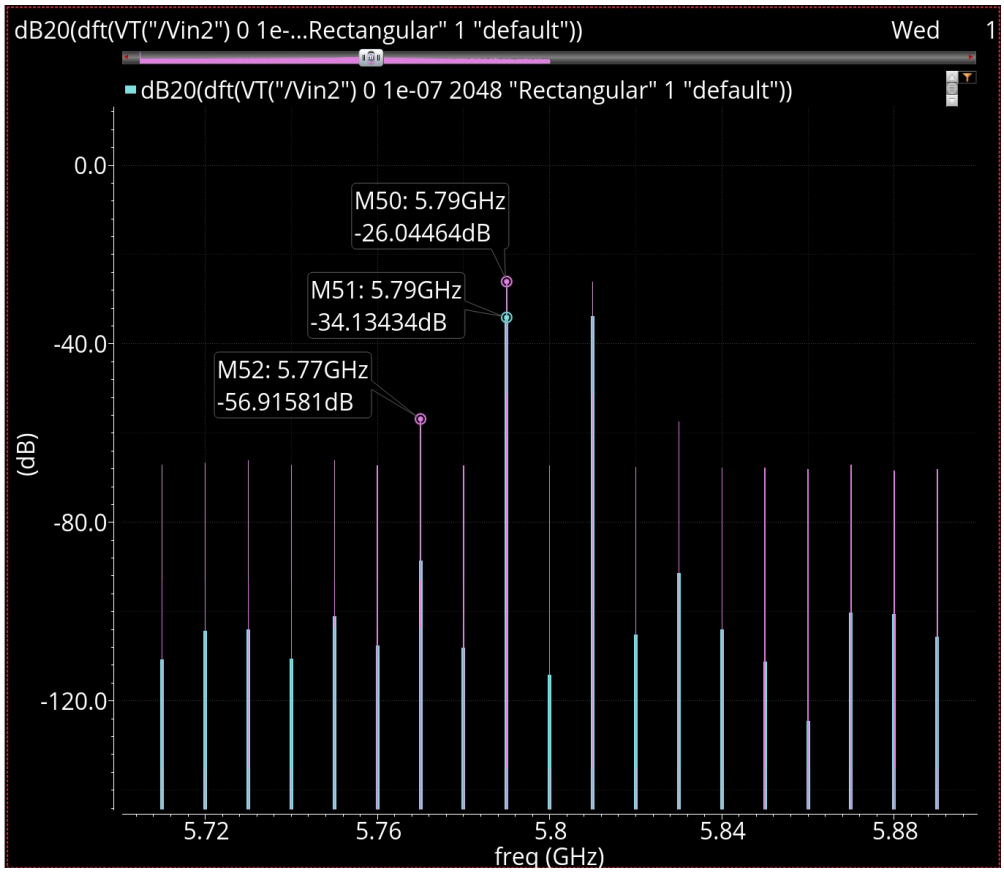
(e). Determine the IIP3 and voltage gain of the stage shown on the right at 5.8 GHz. The role and modeling of L2 are similar to those of L1.

By sweeping the value of L2, the value of it should be 3.9nH for the resonance frequency of 5.8 GHz.



Two tones at 5.79 GHz and 5.81 GHz with 20 mV amplitudes are applied to the input and by performing DFT, the input/output voltage spectrum was plotted as below. All plots are on a dB20 scale.





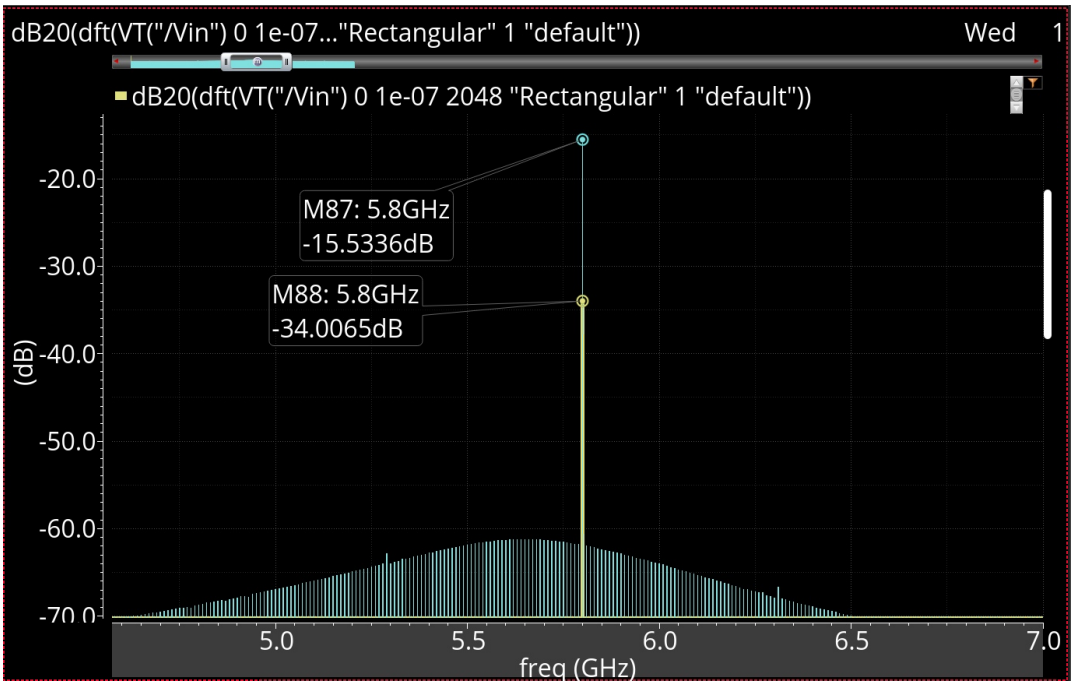
Output: -26.04 dBV = -13.0dBm; Input: 20mV = -21 dBm; IM3: -56.91 dBV = -43.9dBm

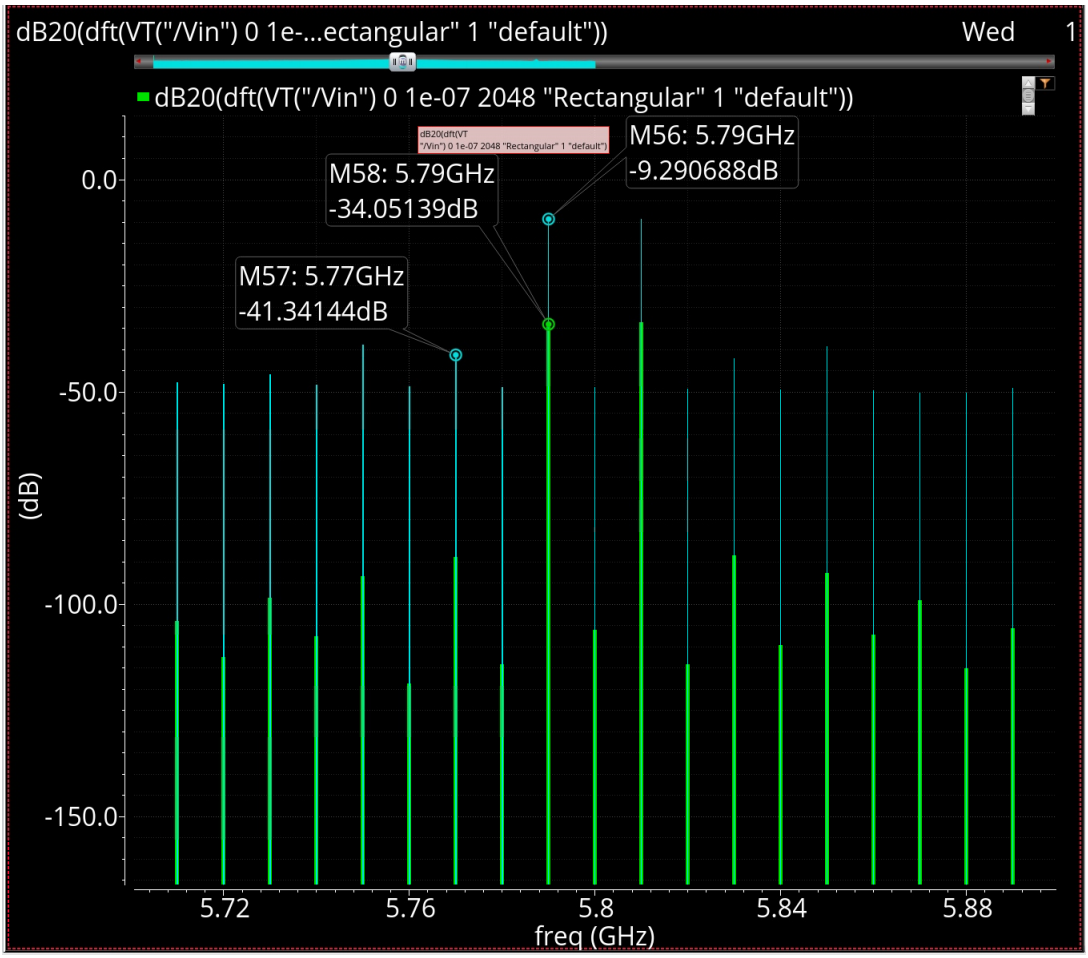
$$G = -27.18dBV - (-34.04dBV) = 6.86\ dB$$

$$IIP3 = \frac{\Delta P}{2} + P_{in} = \frac{-13 + 43.9}{2} - 34 = -18.55\ dBm$$

(f). Now remove M2 from the (a) and place the two stages in a cascade structure, and find the overall voltage gain and IIP3. How closely do these results agree with those obtained from parts (d) and (e) and Eq. (2.67) in the text.

After cascading two stages, two tones at 5.79 GHz and 5.81 GHz with 20 mV amplitudes are applied to the input and by performing DFT, the input/output voltage spectrum was plotted as below. All plots are on a dB20 scale.





Output: -9.30 dBV = 3.7dBm; Input: 20mV = -34 dBV; IM3: -41.34 dBV = -28.3dBm

$$G = -15.53dBV - (-34dBV) = 18.74\ dB$$

$$IIP3 = \frac{\Delta P}{2} + P_{in} = \frac{3.7 + 28.3}{2} - 34 = -18\ dBV$$

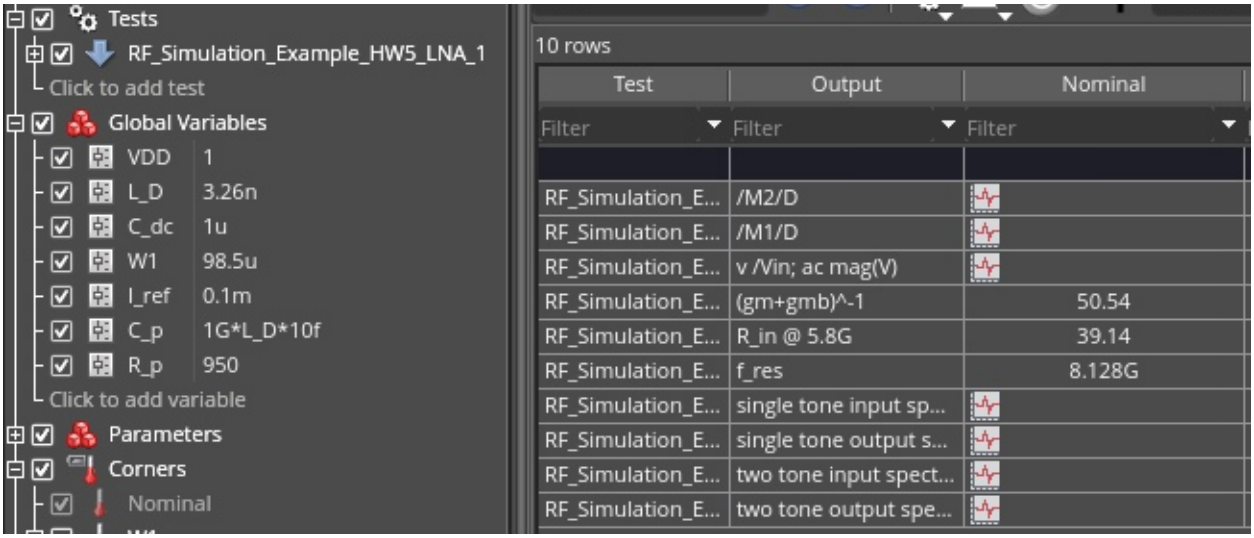
Considering the standalone results of two stages, the calculated gain and IIP3 are:

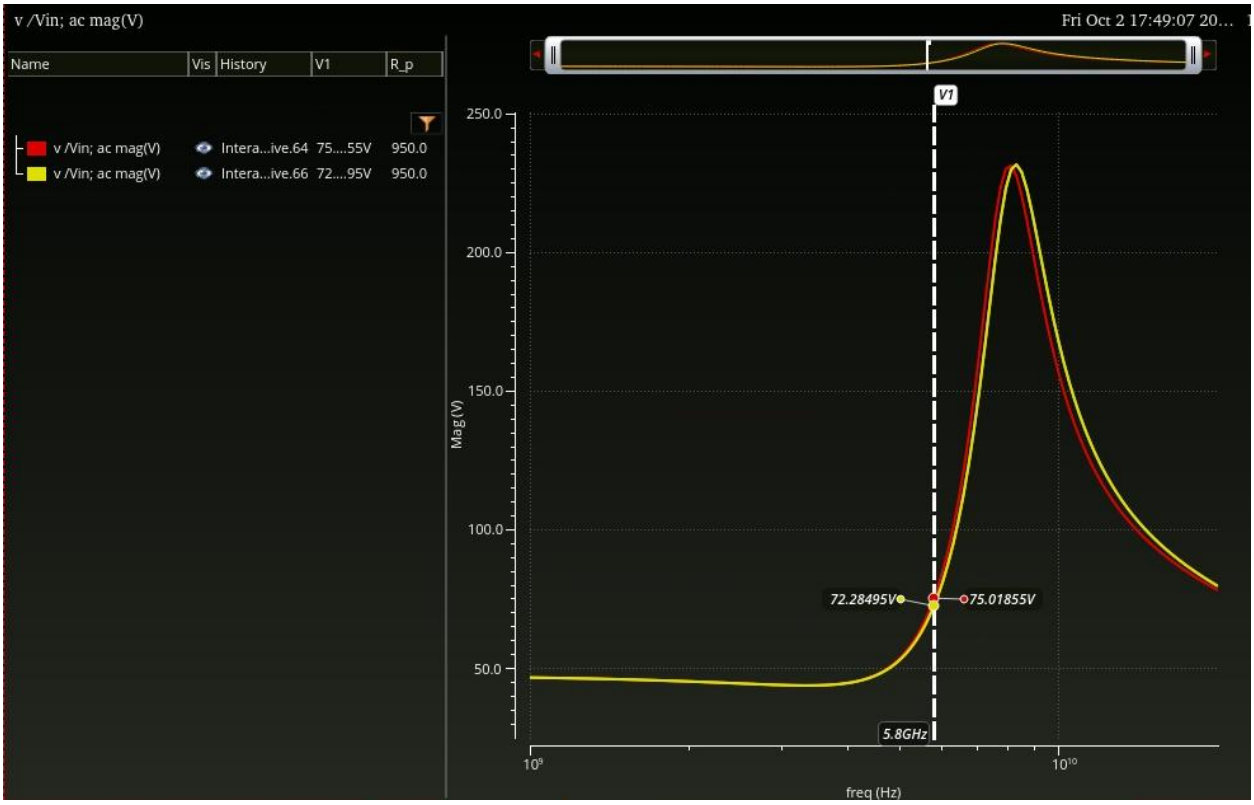
$$G_{calc} = 6.86dB + 9.98dB = 16.84\ dB$$

$$IIP3 = \left(\frac{1}{IIP3{,}1} + \frac{G1}{IIP3{,}2}\right)^{-1} = \left(\frac{1}{(547.68mVpp)^2} + \frac{9.95}{(333.83mVpp)^2}\right)^{-\frac{1}{2}} = -28.63\ dBV$$

The values are different. It is because the cascaded equation is a rough estimation. In real cascading, the second stage is driven by the output of the first stage, yielding IIP3 of the second stage changed.

(g). Does the input resistance of the first stage change when the second stage is added? Why?

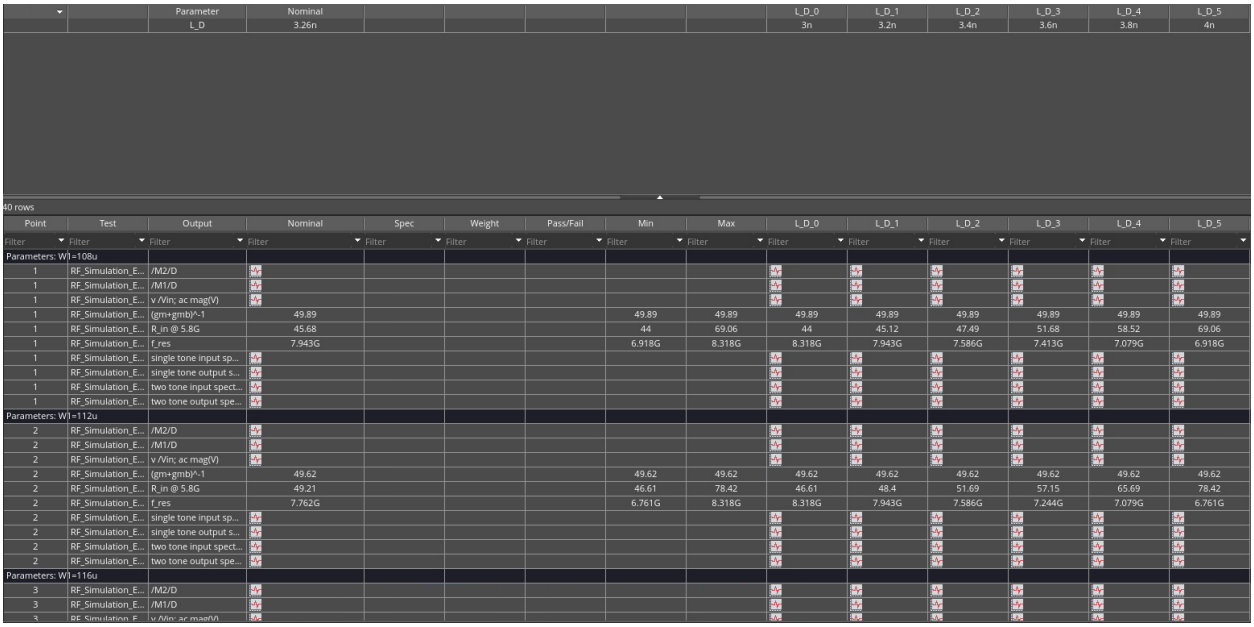




The new input resistance is 39.41 Ohm. Adding the second stage does not change the input impedance much since the loading of the second stage was already modeled. The small change is due to the Cgd of each stage.

* For simultaneous input matching and output tuning, L and W were swept together.

L=3.22nH and W=115u gave the result below.



Parameter	Nominal	L_D_0	L_D_1	L_D_2	L_D_3	L_D_4	L_D_5
L_D	3.26n	3n	3.2n	3.4n	3.6n	3.8n	4n

40 rows

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	L_D_0	L_D_1	L_D_2	L_D_3	L_D_4	L_D_5
Parameters: W1=108u														
1	RF_Simulation_E...	/M2/D												
1	RF_Simulation_E...	/M1/D												
1	RF_Simulation_E...	v /Vin; ac mag(V)												
1	RF_Simulation_E...	(gm+gmb)^-1	49.89				49.89	49.89	49.89	49.89	49.89	49.89	49.89	49.89
1	RF_Simulation_E...	R_in @ 5.8G	45.68				44	69.06	44	45.12	47.49	51.68	58.52	69.06
1	RF_Simulation_E...	f_res	7.943G				6.918G	8.318G	8.318G	7.943G	7.586G	7.413G	7.079G	6.918G
1	RF_Simulation_E...	single tone input sp...												
1	RF_Simulation_E...	single tone output s...												
1	RF_Simulation_E...	two tone input spect...												
1	RF_Simulation_E...	two tone output spe...												
Parameters: W1=112u														
2	RF_Simulation_E...	/M2/D												
2	RF_Simulation_E...	/M1/D												
2	RF_Simulation_E...	v /Vin; ac mag(V)												
2	RF_Simulation_E...	(gm+gmb)^-1	49.62				49.62	49.62	49.62	49.62	49.62	49.62	49.62	49.62
2	RF_Simulation_E...	R_in @ 5.8G	49.21				46.61	78.42	46.61	48.4	51.69	57.15	65.69	78.42
2	RF_Simulation_E...	f_res	7.762G				6.761G	8.318G	8.318G	7.943G	7.586G	7.244G	7.079G	6.761G
2	RF_Simulation_E...	single tone input sp...												
2	RF_Simulation_E...	single tone output s...												
2	RF_Simulation_E...	two tone input spect...												
2	RF_Simulation_E...	two tone output spe...												
Parameters: W1=116u														
3	RF_Simulation_E...	/M2/D												
3	RF_Simulation_E...	/M1/D												
3	RF_Simulation_E...	v /Vin; ac mag(V)												

Test	Output	Nominal
RF_Simulation_E...	/M2/D	
RF_Simulation_E...	/M1/D	
RF_Simulation_E...	v /Vin; ac mag(V)	
RF_Simulation_E...	(gm+gmb)^-1	49.48
RF_Simulation_E...	R_in @ 5.8G	51.43
RF_Simulation_E...	f_res	7.762G
RF_Simulation_E...	single tone input sp...	
RF_Simulation_E...	single tone output s...	
RF_Simulation_E...	two tone input spect...	
RF_Simulation_E...	two tone output spe...	

Test	Output	Nominal
RF_Simulation_E...	/M2/D	
RF_Simulation_E...	/M1/D	
RF_Simulation_E...	v /Vin; ac mag(V)	eval err
RF_Simulation_E...	(gm+gmb)^-1	49.44
RF_Simulation_E...	R_in @ 5.8G	eval err
RF_Simulation_E...	f_res	5.754G
RF_Simulation_E...	single tone input sp...	eval err
RF_Simulation_E...	single tone output s...	
RF_Simulation_E...	two tone input spect...	eval err
RF_Simulation_E...	two tone output spe...	